Magnetosphere and changing fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use evidence to distinguish a magnetic-field model from an unsupported space-weather claim.

SUCCESS CRITERIA - I can...

1. Connect changing magnetic fields with induced electric effects.
2. Distinguish an assumption from an observation.
3. Name evidence required for a space-weather prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “magnetosphere” and “induction” to compare your model or result. Sentence starters are optional.

magnetosphere: a region strongly influenced by a body’s magnetic field

induction: producing an emf through a changing magnetic flux

flux: a measure of magnetic field passing through a chosen area

01 Warm-up: bring back an earlier idea

At fixed current, move from r to 3 r from a long straight wire. Using B proportional to 1/r, compare field strength.

Word help: current = rate of charge flow

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

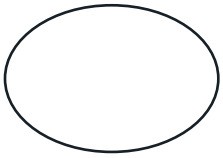
Reference: $|emf_{avg}| = N|\Delta\Phi|/\Delta t$; $\Phi = BA$.
Uniform perpendicular field for $\Phi = BA$. Use supplied/measured field changes; an asteroid alone is not evidence of an electromagnetic pulse.

Worked reasoning: A changing magnetic environment can induce voltages in conducting loops. A static magnetic field alone is not the same as a changing flux through a particular circuit.

Your turn: Sketch a conducting loop and field arrows; show one way flux can change.

Sentence starter: The … in my model represents … . Its label shows … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

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Continue down each column.

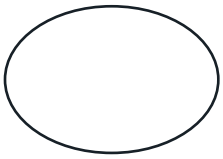
Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw a coil in a changing magnetic field.
Label what changes and where an induced emf could appear.

Sentence starter: The … in my model represents … . Its label shows … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

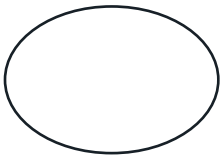
Marzano check - circle: 1 2 3

04 Apply the physics

A coil simulation produces 2 V for one magnetic-field change rate. Double that rate while keeping coil area, orientation and turns fixed.
Predict the new emf and label what changes in the flux model.

Sentence starter: I predict … . My model supports this because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A headline claims an approaching rock will disable power grids through an EMP, but gives no field measurements or mechanism.
Name two pieces of evidence needed before accepting it.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Explain what a potential difference of 6 V means in joules per coulomb.
Label voltage, charge and energy with their distinct units.

Word help: voltage = electric potential difference; energy transferred per charge

Sentence starter: Both … . They differ in … because … .

Initial / supply	Final / component
_____	_____
V: ____ J/C	V: ____ J/C

Charge: ____ C Energy transfer: ____ J

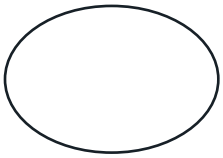
Marzano check - circle: 1 2 3

07 Exit check

Does a constant flux through a stationary coil create sustained induced emf by Faraday’s law?

Sentence starter: I notice … . This supports … because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

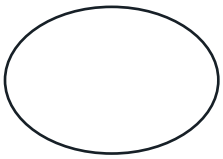
Marzano check - circle: 1 2 3

08 STRETCH: test the model

Compare a small loop and a large loop with the same orientation in the same uniformly changing field.
Predict which may have greater induced emf and state the assumptions.

Sentence starter: Both … . They differ in … because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3